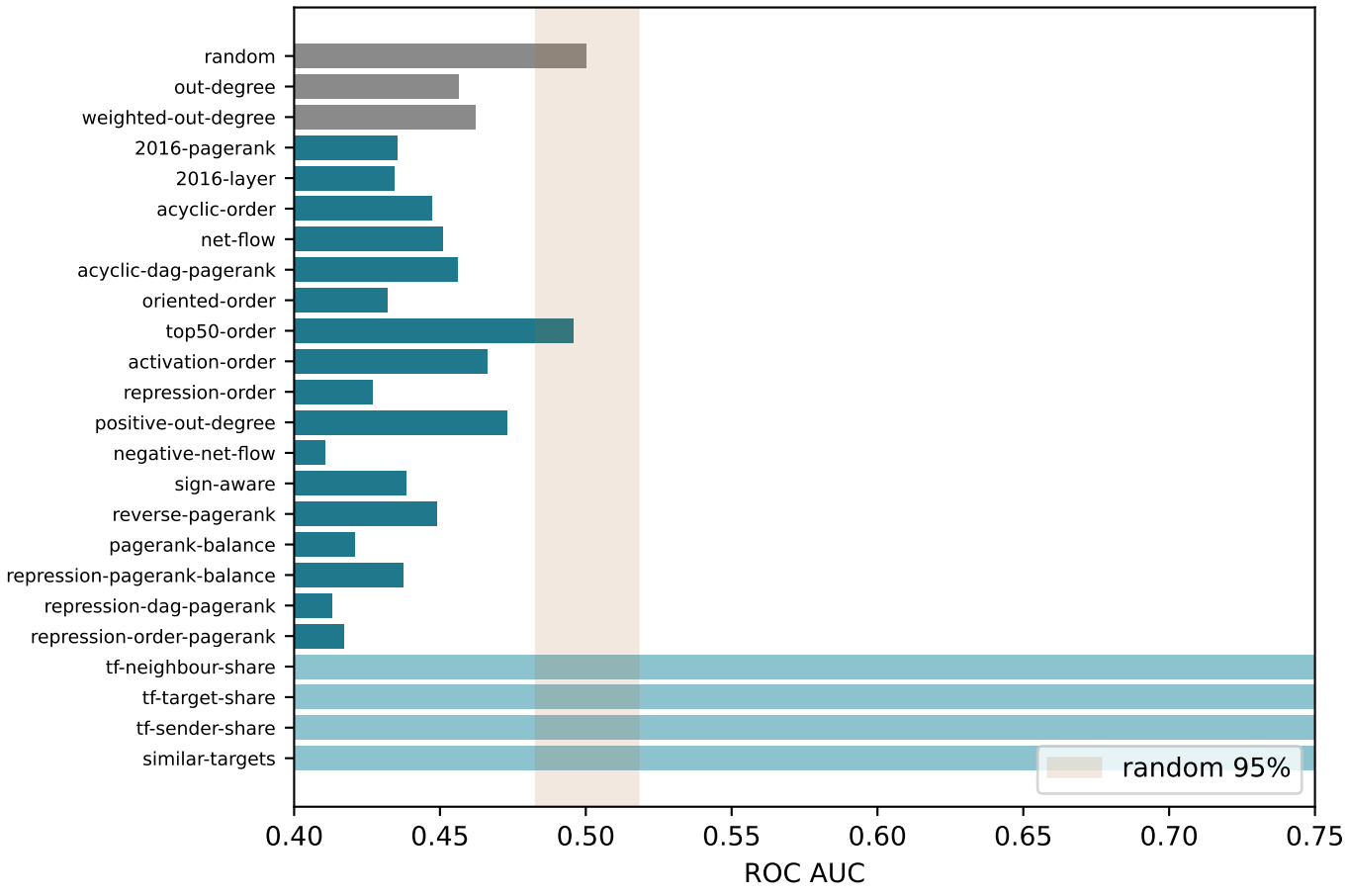
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,974 genes ranked, 1,155 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.6 (4–15)	0.500 (0.483–0.518)				
out-degree	2	0.456	1.000	1.000	-0.006 (-0.007, -0.005)	< 0.001
weighted-out-degree	3	0.462	0.997	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	8	0.435	0.762	1.000	-0.027 (-0.038, -0.014)	< 0.001
2016-layer	4	0.434	0.991	1.000	-0.028 (-0.042, -0.014)	< 0.001
acyclic-order	3	0.447	0.997	1.000	-0.015 (-0.021, -0.009)	< 0.001
net-flow	3	0.451	0.997	1.000	-0.011 (-0.021, -0.002)	0.014
acyclic-dag-pagerank	6	0.456	0.931	1.000	-0.006 (-0.011, -0.001)	0.012
oriented-order	4	0.432	0.991	1.000	-0.030 (-0.041, -0.020)	< 0.001
top50-order	12	0.496	0.241	0.681	+0.034 (+0.017, +0.052)	< 0.001
activation-order	11	0.466	0.360	1.000	+0.004 (-0.003, +0.010)	0.248
repression-order	2	0.427	1.000	1.000	-0.035 (-0.047, -0.024)	< 0.001
positive-out-degree †	8	0.473	0.762	0.998	+0.011 (+0.008, +0.014)	< 0.001
negative-net-flow †	2	0.411	1.000	1.000	-0.051 (-0.065, -0.038)	< 0.001
sign-aware †	3	0.438	0.997	1.000	-0.024 (-0.030, -0.017)	< 0.001
reverse-pagerank †	2	0.449	1.000	1.000	-0.013 (-0.016, -0.011)	< 0.001
pagerank-balance †	1	0.421	1.000	1.000	-0.041 (-0.054, -0.030)	< 0.001
repression-pagerank-balance †	7	0.437	0.863	1.000	-0.025 (-0.037, -0.013)	< 0.001
repression-dag-pagerank †	7	0.413	0.863	1.000	-0.049 (-0.062, -0.037)	< 0.001
repression-order-pagerank †	4	0.417	0.991	1.000	-0.045 (-0.056, -0.033)	< 0.001
tf-neighbour-share ‡	99	0.778	< 0.001	< 0.001	+0.316 (+0.294, +0.336)	< 0.001
tf-target-share ‡	96	0.770	< 0.001	< 0.001	+0.308 (+0.285, +0.330)	< 0.001
tf-sender-share ‡	99	0.760	< 0.001	< 0.001	+0.298 (+0.277, +0.320)	< 0.001
similar-targets ‡	97	0.750	< 0.001	< 0.001	+0.288 (+0.263, +0.311)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,067	0.045 (2.64×)	3.24×	0.76×	0.82 (0.81)
resolve_reciprocals → drop_sinks → prune_triangles	30,796	0.004 (1.04×)	3.72×	0.76×	0.45 (0.41)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	21,969	0.003 (0.89×)	4.32×	0.74×	0.37 (0.34)
absolute	31,067	0.045 (2.64×)			
absolute → minimum_feedback_order	30,827	0.030 (2.25×)			
orient_reciprocals ratio=2.0 → absolute	30,891	0.017 (1.79×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,891	0.017 (1.80×)			
top_outgoing k=50 → absolute	31,067	0.017 (4.89×)			
top_outgoing k=50 → absolute → minimum_feedback_order	30,694	0.012 (4.72×)			
keep_sign sign=1 → absolute	31,046	0.042 (3.24×)			
keep_sign sign=1 → absolute → minimum_feedback_order	30,633	0.027 (2.79×)			
keep_sign sign=-1 → absolute	29,474	0.003 (0.75×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	28,990	0.003 (0.77×)			
keep_sign sign=-1	29,474	0.003 (0.75×)		0.75×	0.24 (0.24)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,647,483 edges (687,696 negative), largest strong component 11,967.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,647,483 edges (687,696 negative), largest strong component 11,967.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 663,537 edges (609,768 negative), largest strong component 11,691.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 663,537 edges (609,768 negative), largest strong component 11,691; DAG keeps 391,309 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,647,483 edges (0 negative), largest strong component 11,967; DAG keeps 2,025,375 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,647,483 edges (0 negative), largest strong component 11,967.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,647,483 edges (0 negative), largest strong component 11,967; DAG keeps 2,025,375 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,503,408 edges (0 negative), largest strong component 10,185; DAG keeps 1,490,941 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 523,111 edges (0 negative), largest strong component 11,963; DAG keeps 370,596 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,959,787 edges (0 negative), largest strong component 11,954; DAG keeps 1,428,089 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 687,696 edges (0 negative), largest strong component 10,877; DAG keeps 602,511 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,647,483 edges (687,696 negative), largest strong component 11,967.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,647,483 edges (687,696 negative), largest strong component 11,967.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,647,483 edges (687,696 negative), largest strong component 11,967.

reverse-pagerank: Highest reverse PageRank (raw graph)